Experiment 1

AIM: Creation of a Data Warehouse.

- **Build Data Warehouse/Data Mart (using open source tools like Pentaho Data Integration Tool, Pentaho Business Analytics; or other data warehouse tools like Microsoft-SSIS, Informatica, Business Objects,etc.,)**
- **Design multi-dimensional data models namely Star, Snowflake and Fact Constellation schemas for any one enterprise (ex. Banking, Insurance, Finance, Healthcare, manufacturing, Automobiles, sales etc).**
- **Write ETL scripts and implement using data warehouse tools.**
- **Perform Various OLAP operations such slice, dice, roll up, drill up and pivot**

Description: Outline and guide for creating a **Data Warehouse**, including ETL implementation, schema design, and OLAP operations, using tools like **Pentaho**, **SSIS**, or **Informatica**.

What is Pentaho Data Integration (PDI)?

Pentaho Data Integration (PDI), also known as **Kettle**, is an open-source tool designed to help you perform **ETL processes** — that is, **Extract**, **Transform**, and **Load** data.

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Breakdown:

- **Extract**: PDI connects to many types of data sources like databases, Excel files, CSVs, XML, JSON, etc., to pull data out.
- **Transform**: It allows you to clean, filter, combine, and modify data. For example, changing date formats, removing duplicates, or applying business rules.
- **Load**: After transformation, PDI loads the data into your target system, like a data warehouse, database, or data mart.

Why Use PDI?

- **Visual, drag-and-drop interface**: You build data workflows visually without writing complex code.
- **Supports many data sources and targets**.
- **Scalable & powerful**: Can handle simple and complex data pipelines.
- **Open-source & free**, but also available with enterprise support from Hitachi Vantara.

Example Use Case:

You have patient data stored in multiple Excel files and a hospital database. Using PDI, you can:

- Extract patient info from all sources.
- Clean the data (fix formats, remove errors).
- Merge it into one unified table in your data warehouse.

Summary:

Term	Meaning
Pentaho Data Integration (PDI)	Tool for ETL workflows
ETL	Extract, Transform, Load
Goal	Move & clean data for analysis

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Detailed **notes** on each component of the *Creation of a Data Warehouse* project, broken down into major sections.

🗀 1. Build Data Warehouse/Data Mart

Tools (Choose One):

- **Open-source tools:**
  - *Pentaho Data Integration (PDI)* – for ETL processes.
  - *Pentaho Business Analytics* – for reporting and dashboards.
- **Proprietary tools:**
  - *Microsoft SSIS* (SQL Server Integration Services)
  - *Informatica PowerCenter*
  - *SAP Business Objects*

❖ What is a Data Warehouse?

- A **Data Warehouse** is a central repository where data from multiple sources is integrated, transformed, and stored to support decision-making and analysis.

➢ **What is a Data Mart?**

A **Data Mart** is a subset of a data warehouse, focused on a specific business area (e.g., sales, marketing).

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2. Design Multi-Dimensional Data Models

Select any one enterprise domain. Example: **Healthcare**

Schema Types:

☆ **Star Schema:**

- **Central Fact Table** (e.g., Fact_Patient_Visits)
- **Surrounding Dimension Tables** (e.g., Dim_Patient, Dim_Doctor, Dim_Hospital, Dim_Date)

❄ **Snowflake Schema:**

- Extension of Star Schema by **normalizing dimension tables**.
- Example: Dim_Hospital → Dim_Hospital & Dim_Location

Fact Constellation Schema:

- Also called **Galaxy Schema**.
- Contains **multiple fact tables** that share dimension tables.
- Example:
    - Fact_Patient_Visits, Fact_Medication_Records
    - Shared Dimensions: Dim_Patient, Dim_Date

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3. Write ETL Scripts

ETL = Extract, Transform, Load

1. **Extract**
    - Retrieve data from multiple sources (CSV, Excel, RDBMS, APIs)
    - Example:
    - SELECT * FROM hospital_db.patients;
2. **Transform**
    - Clean, standardize, and apply business rules.
    - Example: Convert gender M/F → Male/Female

3. **Load**
    - Load the data into the data warehouse fact and dimension tables.
    - Tools like **Pentaho PDI**, **SSIS**, or **Informatica** help in drag-and-drop ETL workflows.

4. OLAP Operations

OLAP = **Online Analytical Processing** – supports multi-dimensional queries.

Common OLAP Operations:

Operation	Description	Example
Slice	Select a single dimension from a cube	View only 2023 patient visits
Dice	Select two or more dimensions	View patient visits for Jan-Mar in 3 hospitals
Roll-up	Aggregation (e.g., Day → Month → Year)	Total visits per quarter
Drill-down	Detailed data (e.g., Year → Month → Day)	View visits on a specific day
Pivot	Rotate data axes to see from different perspectives	Rows as hospitals, columns as months

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Conclusion

The creation of a Data Warehouse involves:

1. Choosing appropriate tools.
2. Designing effective multi-dimensional models (Star, Snowflake, Fact Constellation).
3. Writing robust ETL processes to populate the warehouse.
4. Performing OLAP operations to support analytics.

You can tailor this framework to a domain like **Healthcare, Finance, Retail**, etc., for your project work.

Example with Healthcare data model + ETL flow:

Let's walk through a **complete example** of creating a **Data Warehouse** for a

Healthcare domain, covering all key components:

Use Case: Healthcare - Patient Visit Analytics

Goal: Analyze patient visits across hospitals to identify trends, costs, and doctor performance.

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1. Tool Used: *Pentaho Data Integration (PDI – a.k.a. Kettle)*

- **Reason**: It's open-source, supports drag-and-drop ETL, and integrates easily with MySQL or PostgreSQL data warehouses.

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2. Data Warehouse Schema Design

We'll use a **Star Schema** for simplicity.

Fact Table:

- **Fact_Patient_Visit**
  - Visit_ID (PK)
  - Patient_ID (FK)
  - Doctor_ID (FK)
  - Hospital_ID (FK)
  - Date_ID (FK)
  - Diagnosis
  - Treatment_Cost
  - Visit_Duration_Minutes

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Dimension Tables:

Dim_Patient

- Patient_ID (PK)
- Name
- Gender
- Age
- City

Dim_Doctor

- Doctor_ID (PK)
- Name

- Specialization
- Experience_Years

Dim_Hospital

- Hospital_ID (PK)
- Hospital_Name
- Location
- Type (e.g., General, Specialty)

Dim_Date

- Date_ID (PK)
- Date
- Month
- Quarter
- Year

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3. ETL Process Using Pentaho

Extract:

- Connect to source systems (e.g., hospital Excel sheet, MySQL patient database)
- Example: Import from CSV of patient visits.

Transform:

- Convert date formats (e.g., DD-MM-YYYY → YYYY-MM-DD)
- Clean null/missing data
- Map M/F → Male/Female in Patient data
- Lookup or generate surrogate keys for dimensions

Load:

- Load Dim_Patient, Dim_Doctor, Dim_Hospital, Dim_Date
- Load Fact_Patient_Visit with foreign keys referencing dimension tables

Example PDI Job:

- CSV Input → Select Values → Lookup Dimension ID → Insert into Fact Table

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4. OLAP Operations

Assuming OLAP cube created on this star schema:

✓ **Slice:**

- **Query**: Show all visits in the year 2024
- **Effect**: Filters data to a single time slice (Year = 2024)

✓ **Dice:**

- **Query**: Patients from Mumbai treated by Cardiologists in 2024
- **Effect**: Filters on multiple dimensions

✓ **Roll-up:**

- **Query**: Group visits by Quarter
- **Effect**: Aggregates data from month to quarter level

✓ **Drill-down:**

- **Query**: From quarterly stats → drill to monthly visits
- **Effect**: View more granular data

✓ **Pivot:**

- **Query**: Pivot table with Doctors as rows and Hospital as columns
- **Effect**: Rotate cube axes for new view

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Example Query (for OLAP Cube or Report Tool)

```
SELECT d.Specialization, h.Hospital_Name, COUNT(*) AS Visit_Count FROM
Fact_Patient_Visit f
JOIN Dim_Doctor d ON f.Doctor_ID = d.Doctor_ID JOIN
Dim_Hospital h ON f.Hospital_ID = h.Hospital_ID JOIN
Dim_Date dt ON f.Date_ID = dt.Date_ID  WHERE dt.Year =
2024
GROUP BY d.Specialization, h.Hospital_Name;
```

Output:

Specialization	Hospital_Name	Visit_Count
Cardiology	Fortis Hospital	1200
Neurology	Apollo Hospitals	850

✓ **Summary**

Step	Description
Tool	Pentaho PDI for ETL
Schema	Star Schema
Fact Table	Patient Visit
Dimensions	Patient, Doctor, Hospital, Date
OLAP Ops	Slice, Dice, Roll-up, Drill-down, Pivot
Result	Analytical insights into healthcare service delivery

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